TOPIC 01: SIMPLE INTEREST & LINEAR TIME VALUE

Formal Mathematical Foundations, Real Analysis Bounds, and Hand-Solved Case Studies

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1. Mathematical Definition & Axiomatic Assumptions

Simple Interest represents the foundational linear model in financial valuation. It assumes that interest earned across any discrete time increment is *never reinvested* into the principal base.

Let the initial principal be $P > 0$, nominal annual interest rate be $r \geq 0$ (decimal fraction: $r = R / 100$ for percentage rate $R\%$), and investment horizon be $t \geq 0$ years.

$$I(t) = P \cdot r \cdot t$$

$$A(t) = P + I(t) = P(1 + r \cdot t)$$

where $I(t)$ is accrued interest and $A(t)$ is terminal accumulated capital.

2. Mathematical Derivation & Differential Formulation

Axiomatic Derivation: Under simple accumulation, interest I satisfies three simultaneous linear proportionalities: (1) $I \propto P$ (scale invariance), (2) $I \propto r$ (yield homogeneity), and (3) $I \propto t$ (temporal linearity). Combining these yields $I = k \cdot P \cdot r \cdot t$, with $k = 1$ by convention.

Differential Equation Perspective: Examining instantaneous wealth growth $A(t)$ reveals the key distinction between linear and compound regimes:

Model	Governing Differential Equation	Solution Function $A(t)$
Simple Interest	$dA/dt = P \cdot r$ (constant rate)	$A(t) = P(1 + r \cdot t)$ [Linear]
Continuous Compounding	$dA/dt = r \cdot A(t)$ (wealth-dependent)	$A(t) = P \cdot e^{rt}$ [Exponential]

Real Analysis Theorem (Bernoulli Inequality Application):
For any annual rate $r > 0$ and fractional time period $t \in (0, 1)$, simple interest is *strictly superior* to annual compound interest:

$$\mathbf{(1 + r)^t < 1 + r \cdot t \quad \forall\, t \in (0, 1)}$$

Proof: Define $f(t) = (1 + r)^t$. The second derivative $f''(t) = [\ln(1 + r)]^2(1 + r)^t > 0$, establishing strict convexity on $[0, 1]$. The linear chord connecting $(0, 1)$ and $(1, 1+r)$ is $g(t) = 1 + rt$. By strict convexity, the curve $f(t)$ lies strictly beneath its chord $g(t)$ on the interior $(0, 1)$.

3. Worked Example 1: Standard Multi-Year Bond Yield (Hand Solved)

Problem: An investor places P = INR 100,000 in a sovereign debt instrument at an annual simple interest rate of $R = 7.50\%$ for $t = 3$ years. Compute total interest accrued and final maturity amount.

Formal Step-by-Step Hand Solution:

- **Step 1: Parameter Identification:** $P = 100{,}000$, $r = 7.50 / 100 = 0.075$, $t = 3.0$ years.
- **Step 2: Interest Accrual:**
  $I = P \cdot r \cdot t = 100{,}000 \times 0.075 \times 3 = 100{,}000 \times 0.225 =$ **INR 22,500.00**.
- **Step 3: Terminal Capital:**
  $A = P + I = 100{,}000 + 22{,}500 =$ **INR 122,500.00**.
- **Step 4: Annual Periodic Verification:**
  Year 1: I_1 = INR 7,500.00 | Year 2: I_2 = INR 7,500.00 | Year 3: I_3 = INR 7,500.00.
  Total = 3 × 7,500 = INR 22,500.00 (verified exact linear accumulation).

4. Worked Example 2: Fractional Tenors & The Banker's Rule (Hand Solved)

Problem: A short-term bridge loan of P = INR 250,000 is issued on **March 15** at an annual simple interest rate of $R = 9.20\%$, and settled on **October 28** of the same non-leap year. Compare interest under:
Method A: Exact / Actual/365 Convention (Sovereign bonds) | **Method B:** Banker's Rule / Actual/360 Convention (Money markets)

Formal Step-by-Step Hand Solution:

- **Step 1: Compute Exact Elapsed Days (*d*):**
  March: (31 – 15) = 16 | Apr: 30 | May: 31 | Jun: 30 | Jul: 31 | Aug: 31 | Sep: 30 | Oct: 28.
  Total elapsed days: d = 16 + 30 + 31 + 30 + 31 + 31 + 30 + 28 = **227 days**.
- **Step 2: Method A (Actual/365 Convention):**
  Time fraction t_{365} = 227 / 365 ≈ 0.6219178 years.
  I_{365} = 250,000 × 0.092 × (227 / 365) = 23,000 × 0.6219178 = **INR 14,304.11**.
  Maturity payment: A_{365} = 250,000 + 14,304.11 = **INR 264,304.11**.
- **Step 3: Method B (Banker's Rule — Actual/360 Convention):**
  Time fraction t_{360} = 227 / 360 ≈ 0.6305556 years.
  I_{360} = 250,000 × 0.092 × (227 / 360) = 23,000 × 0.6305556 = **INR 14,502.78**.
  Maturity payment: A_{360} = 250,000 + 14,502.78 = **INR 264,502.78**.
- **Yield Discrepancy Analysis:**
  $\Delta I = I_{360} - I_{365}$ = 14,502.78 – 14,304.11 = **INR 198.67** excess interest.
  Mathematical insight: Actual/360 artificially scales the effective interest rate by 365 / 360 ≈ 1.01389 (+1.39% yield boost), illustrating why day-count definitions are contractual prerequisites in fixed-income mathematics.

5. Algorithmic Implementation in Python

```
def simple_interest(principal: float, annual_rate: float, time_years: float) -> float:
"""
Computes accrued simple interest: I = P * r * t.
Enforces domain constraints: principal >= 0, annual_rate >= 0, time_years >= 0.
"""
if principal < 0 or annual_rate < 0 or time_years < 0:
raise ValueError("Financial parameters must be non-negative.")
return float(principal * annual_rate * time_years)

def simple_interest_amount(principal: float, annual_rate: float, time_years: float) -> float:
"""Computes terminal accumulated wealth: A = P * (1 + r * t)."""
return float(principal + simple_interest(principal, annual_rate, time_years))
```

6. Avishkar Jury & MSc Admissions Defense Q&A;

Q1: Why is Simple Interest unsuited for modeling multi-period long-term investments?

Defense: Simple interest violates the principle of temporal financial equivalence. In financial mathematics, money received at t = 1 has an opportunity cost; reinvesting that interest should yield further returns. By restricting the growth rate $dA/dt = Pr$ to a constant, simple interest ignores the compounding of reinvested gains, underestimating long-term wealth accumulation exponentially.

Q2: Under what conditions does Simple Interest yield more than Compound Interest?

Defense: By the strict convexity of the exponential function and Bernoulli's inequality, for any positive rate $r > 0$ and fractional time period $t \in (0, 1)$, simple interest accumulation $1 + rt$ strictly exceeds annual compounding $(1 + r)^t$. They intersect at exactly two points: $t = 0$ where both equal 1, and $t = 1$ where both equal $1 + r$.

Q3: What role does Simple Interest play in institutional Quantitative Finance today?

Defense: While rarely used for multi-year retail loans, simple interest remains the global market standard for *money market instruments* with tenors under one year (e.g., Treasury Bills, Commercial Paper, and Interbank repo rates) governed by exact day-count conventions such as Actual/360 and Actual/365.